

Surya Kannan

669-298-4575 | suryakannan913@berkeley.edu | [linkedin.com/in/surya-kannan-9bb9312b0/](https://www.linkedin.com/in/surya-kannan-9bb9312b0/) | github.com/suryakannan913

EDUCATION

University of California Berkeley

College of Computing, Data Science and Society

Berkeley, CA

Aug. 2025 – May 2027

DeAnza College | High School Dual Enrollment

Intersegmental General Education Transfer Curriculum (IGETC)

Cupertino, CA

Aug. 2022 – June 2025

PROJECTS

Course Recommendation Project | *Python, Neo4j, Cypher, FAISS, OpenAI API, Flask* Jan. 2026 – April 2026

- **Engineered a Hybrid RAG System** using **Neo4j (Knowledge Graph)** and **FAISS (Vector)** to compare retrieval accuracy on complex datasets.
- **Achieved 100% accuracy** on multi-hop logical queries using **Cypher**, where standard Vector RAG failed.
- **Boosted Faithfulness scores by 40%** by utilizing Knowledge Graph structures to eliminate LLM hallucinations and grounding errors in GPT-4o responses.

Everguide Wealth Management | *Fastino API, Linkup, Gemini API, Meta Llama* May 2018 – May 2020

- Built a **multi-agentic modular** financial advisory AI platform in **Python** using **FastAPI**, for **portfolio analysis**, **risk assessment**, and **tax optimization**.
- Leveraged **Google Gemini LLM** for intelligent, context-aware recommendations and automated **report generation**, enhancing **decision support** and **client personalization**.
- Engineered a unified **memory management system** with semantic, episodic, and procedural layers, backed by **MongoDB** and **Fastino API** for robust **data storage** and **retrieval**.

TECHNICAL SKILLS

Languages: Python, SQL (BigQuery/PostgreSQL), Java, C/C++, JavaScript

Data Platforms: Google Cloud Platform (GCP), BigQuery, Neo4j, MongoDB

Tools & Frameworks: Flask, FastAPI, Docker, Git, Monte Carlo (Exposure), FAISS

Concepts: Data Observability, ETL Pipelines, Data Quality, Multi-agent AI, SDLC

CERTIFICATIONS

Neo4j - Certified Professional | *Cypher, Schema Design, Labeled Property Graph* March 2026

- **Architected High-Performance Graph Databases** by implementing **advanced property graph models** with **optimized node labels**, relationship types, and indexing strategies.
- **Engineered Complex Cypher Queries** utilizing subqueries, pattern comprehension, and list-processing functions to **extract deep insights** from highly interconnected datasets.
- **Managed Enterprise Graph Operations** by configuring database security, implementing ACID-compliant transactions, and utilizing the APOC library for **advanced data procedures**.

Amazon - Introduction to Software Development | *Python, Java, Javascript* October 2025

- **Managed full-cycle SDLC** by transitioning projects from initial requirements gathering through architecture design and automated testing phases.
- **Streamlined CI/CD pipelines** by implementing automated testing and version control workflows to maintain high deployment standards.

Google - IT Automation with Python | *Python, Bash Scripting, Regex, Git* September 2024

- **Automated System Administration** by developing Python scripts to manage files, and process log data.
- **Implemented Configuration Management** maintaining consistent state across a fleet of virtual machines.

EXPERIENCE

The Coderschool

Python Tutor and mentor

Feb. 2025 – June 2025

Cupertino, CA

Momentum Tennis

Lead Jr. Coach

March 2024 – Aug. 2025

Cupertino, CA